Cairo University
Faculty of Engineering
Department of Computer Engineering

Baligh

Arabic Writing Assistant

A Graduation Project Report Submitted

to

Faculty of Engineering, Cairo University

in Partial Fulfillment of the requirements of the degree

of

Bachelor of Science in Computer Engineering.

Presented by

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Abstract

Write a clear and brief abstract. It should cover: the addressed problem, objectives, followed approach, project outputs, development/testing tools, testing summary, and sponsors if any.

Baligh is an AI-powered Arabic Writing Assistant designed to improve Modern Standard Arabic writing through real-time grammatical error detection, correction, spelling correction, next-word prediction, and explanatory feedback. This section should be replaced by the final project abstract.

الملخص

يُكتب هنا نفس محتوى الـ Abstract ولكن باللغة العربية.

بليغ هو مساعد كتابة عربي مدعوم بالذكاء الاصطناعي يهدف إلى تحسين الكتابة باللغة العربية الفصحى من خلال اكتشاف الأخطاء وتصحيحها واقتراح الكلمات التالية وتقديم تفسيرات تساعد المستخدم على فهم سبب الخطأ وطريقة التصحيح. تُستبدل هذه الفقرة بالملخص العربي النهائي للمشروع.

Acknowledgment

Write your acknowledgment and dedication here.

Contents

List of Figures

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List of Abbreviations

Put abbreviations in alphabetical order.

Abbreviation	**Meaning**
AI	Artificial Intelligence
GEC	Grammatical Error Correction
GED	Grammatical Error Detection
MSA	Modern Standard Arabic
NLP	Natural Language Processing
NWP	Next-Word Prediction

List of Symbols

Put symbols in alphabetical order. Greek symbols should come first, followed by English symbols.

Symbol	**Meaning**
α	Learning rate or weighting coefficient
P	Probability
T	Time or sequence length

Contacts

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Chapter 1: Introduction

This chapter introduces the project, justifies the need for it, states the project outcomes, and explains the organization of the report.

1.1. Motivation and Justification

Explain why the project is needed, what problem it addresses, why the problem is significant, and who benefits from the project.

Arabic writing tools often provide corrections without explaining the grammatical reasoning behind them. Baligh focuses not only on correction, but also on understandable explanations that help users improve their writing over time.

1.2. Project Objectives and Problem Definition

State the project objectives clearly and end with a formal problem definition.

1.3. Project Outcomes

State the final project outcomes: models, datasets, applications, APIs, experiments, reports, or other deliverables.

1.4. Document Organization

Briefly describe the following chapters.

Chapter 2: Market Visibility Study

This chapter surveys the market related to the project and discusses similar tools or platforms. It should explain their limitations and justify the need for Baligh.

2.1. Targeted Customers

Mention the intended customers/users and how they benefit from the project.

2.2. Market Survey

List competing or related products. Discuss each one in a subsection and explain its pros and cons.

2.2.1. Competitive Product 1

Describe the product, strengths, weaknesses, and relevance to Baligh.

2.2.2. Competitive Product 2

Describe the product, strengths, weaknesses, and relevance to Baligh.

2.3. Business Case and Financial Analysis

Describe the possible business case, expected users, pricing assumptions, CapEx, OpEx, revenue, cash flow, and break-even point.

Table 2.1: *Sample Five-Year Financial Projection*

Year	Expected Users	Revenue	Expenses	Profit Before Tax
1	–	–	–	–
2	–	–	–	–
3	–	–	–	–
4	–	–	–	–
5	–	–	–	–

Chapter 3: Literature Survey

This chapter should include necessary engineering and non-engineering background, followed by a concise literature review of recent work related to the project.

3.1. Background on Arabic NLP

Explain the most important concepts needed to understand the project, such as morphology, spelling correction, grammatical error correction, and language modeling.

3.2. Background on Writing Assistance Systems

Explain writing assistants, correction pipelines, feedback generation, and user-facing explanations.

3.3. Comparative Study of Previous Work

Classify and compare previous work. Focus on relevance, strengths, limitations, datasets, methods, and results.

Table 3.1: *Comparison of Related Arabic Writing Assistance Approaches*

Work/System	Main Approach	Strengths	Limitations
System 1	Rule-based	Explainable	Limited coverage
System 2	Neural	Strong accuracy	Less interpretable
System 3	Hybrid	Balanced	Requires integration effort

3.4. Implemented Approach

State the approach chosen from the reviewed approaches and justify why it was selected. Mention any modifications added by the team.

Chapter 4: System Design and Architecture

This chapter is the main body of the report. It should answer: what has been done, and how has it been done? The discussion should flow from the overall architecture to modules and submodules.

4.1. Overview and Assumptions

Introduce the system design and list the assumptions used during design and implementation.

4.2. System Architecture

Describe the overall architecture and how modules communicate.

4.2.1. Block Diagram

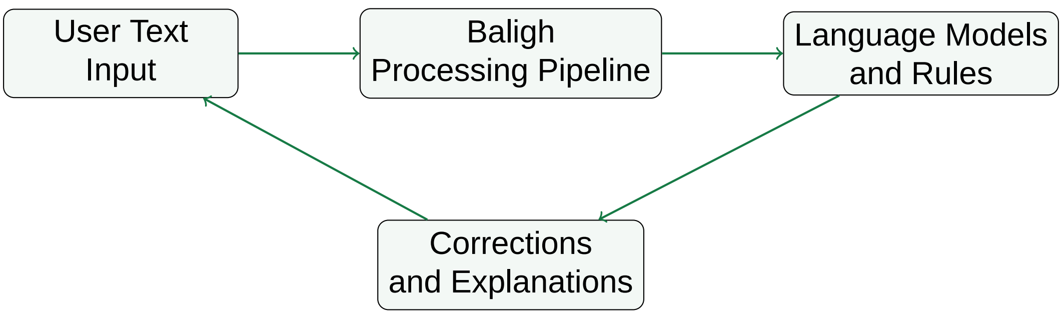

Figure 4.1: *High-level architecture of the Baligh system.*

4.3. Module 1: Grammatical Error Detection

4.3.1. Functional Description

Explain what this module does.

4.3.2. Modular Decomposition

Break the module into submodules.

4.3.3. Design Constraints

Explain constraints such as latency, Arabic ambiguity, model size, and data availability.

4.3.4. Other Description of Module 1

Add any other relevant technical details.

4.4. Module 2: Correction and Explanation Generation

4.4.1. Functional Description

Explain correction generation and explanation generation.

4.4.2. Modular Decomposition

Break the module into submodules.

4.4.3. Design Constraints

Explain constraints such as explanation quality, grammatical accuracy, and user readability.

4.4.4. Other Description of Module 2

Add any other relevant technical details.

Chapter 5: System Testing and Verification

This chapter describes how the system was tested and verified.

5.1. Testing Setup

Describe hardware, software, datasets, versions, and test environment.

5.2. Testing Plan and Strategy

Describe unit tests, integration tests, user testing, model evaluation, and manual review.

5.2.1. Module Testing

Explain how each module was tested independently.

5.2.2. Integration Testing

Explain how the full pipeline was tested after integration.

5.3. Test Schedule

Add the timeline/schedule of testing activities.

5.4. Comparative Results to Previous Work

Compare your results with baselines or previous work where applicable.

Chapter 6: Conclusions and Future Work

6.1. Faced Challenges

Discuss technical, management, research, data, and integration challenges.

6.2. Gained Experience

Discuss technical and non-technical learning outcomes.

6.3. Conclusions

Summarize what the project achieved and why it matters.

6.4. Future Work

Mention possible future improvements and extensions.

References

[1] Author Name, “Paper or Book Title,” *Conference or Journal Name*, Year.

[2] Author Name, “Another Reference Title,” Publisher or Venue, Year.

Chapter A: Development Platforms and Tools

List programming languages, frameworks, libraries, operating systems, hardware, datasets, and development tools.

Chapter B: Use Cases

Describe system use cases, actors, preconditions, main flows, alternative flows, and expected results.

Chapter C: User Guide

This appendix should be included. Explain how to install, run, and use the final system. Add screenshots or step-by-step instructions when available.

Chapter D: Code Documentation

Document important modules, classes, APIs, configuration files, and code structure.

Chapter E: Feasibility Study

This appendix should be included. Add market, technical, operational, and financial feasibility details.